The determinant of a two by two matrix is the product of the diagonal entries minus the product of the others:

$$\det A = \begin{vmatrix} 7 & -1 \\ 11 & 3 \end{vmatrix} = 32$$

Expanding along the first row gives the determinant of a three by three matrix:

$$\det B = \begin{vmatrix} 2 & 1 & 7 \\ 7 & 1 & 7 \\ -3 & -7 & -9 \end{vmatrix} = -200$$

Scaling entries scales the terms of the determinant:

$$\begin{vmatrix} 9a & b \\ 3c & d \end{vmatrix} = 9ad - 3bc$$

The characteristic condition below is a quadratic equation in x:

$$\begin{vmatrix} x-7 & 9 \\ 1 & x+7 \end{vmatrix} = 0$$

Cramer's rule writes each unknown as a ratio of determinants:

$$x = \frac{\begin{vmatrix} 2 & 11 \\ 1 & -3 \end{vmatrix}}{\begin{vmatrix} 15 & 11 \\ -9 & -3 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} 15 & 2 \\ -9 & 1 \end{vmatrix}}{\begin{vmatrix} 15 & 11 \\ -9 & -3 \end{vmatrix}}$$

Matrices

Matrices record data, transformations and systems of equations. Brackets and parentheses are both common. Transposing a matrix swaps its rows and columns:

$$A = \begin{bmatrix} -7 & -7 & -4 \\ -1 & 11 & 7 \end{bmatrix}, \qquad A^T = \begin{bmatrix} -7 & -1 \\ -7 & 11 \\ -4 & 7 \end{bmatrix}$$

A matrix times a column vector gives another column vector:

$$\begin{pmatrix} 2 & -7 \\ 1 & -9 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

The inverse of a two by two matrix uses the determinant and swaps entries:

$$A^{-1} = \frac{1}{15} \begin{bmatrix} 3 & -1 \\ 9 & 2 \end{bmatrix}$$

The augmented matrix of a system keeps the right-hand side beside the coefficients:

$$\left[\begin{array}{cc|c} -5 & 0 & 1 \\ -4 & -7 & 2 \end{array}\right]$$